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Tsi12C Tenkasi District

Common Quarterly Examination - September 2026

25-09-2026.

Standard 12

CHEMISTRY

Time Allowed: 3.00 Hours Maximum Marks: 70

PART - I

I. Choose the best answer: **15×1=15**

1) Electrochemical process is used to extract
   a) Iron b) Lead c) Silver d) Sodium

2) If 'a' stands for the edge length of cubic system: sc, bcc and fcc. Then the ratio of radii of spheres in these systems will be respectively.
   a) $\left(\frac{1}{2}a : \frac{\sqrt{3}}{2}a : \frac{\sqrt{2}}{2}a\right)$ b) $(\sqrt{2}a : \sqrt{3}a : \sqrt{2}a)$
   c) $\left(\frac{1}{2}a : \frac{\sqrt{3}}{4}a : \frac{1}{2\sqrt{2}}a\right)$ d) $\left(\frac{1}{2}a : \sqrt{3}a : \frac{1}{\sqrt{2}}a\right)$

3) **Assertion** : Phenol is more reactive than benzene towards electrophilic substitution reaction.
   Reason : In the case of phenol, the intermediate uranium ion is more stabilized by resonance.
   a) If both assertion and reason are true and reason is the correct explanation of assertion.
   b) If both assertion and reason are true but reason is not the correct explanation of assertion.
   c) Assertion is true but reason is false.
   d) Both assertion and reason are false.

4) The element that does not show catenation among the following p-block element is
   a) Carbon b) Silicon c) Lead d) Germanium

5) In a first order reaction x→y : if K is the rate constant and the initial concatenation of the reactant x is 0.1M, then the half life is
   a) $\left(\frac{\log 2}{K}\right)$ b) $\frac{0.693}{(0.1)K}$ c) $\left(\frac{\ell n_2}{K}\right)$ d) None of these

6) Benzoic Acid $\xrightarrow[(ii)\ \Delta]{(i)\ NH_3}$ A $\xrightarrow{NaOBr}$ B $\xrightarrow{NaNO_2/HCl}$ C "C" is
   a) anilinum chloride b) O - nitro aniline
   c) benzene diazonium chloride d) m - nitro benzoic acid

7) Which one of the following orders is acid correct for the bond dissociation enthalpy of halogen molecules?
   a) $Br_2 > I_2 > F_2 > Cl_2$ b) $F_2 > Cl_2 > Br_2 > I_2$
   c) $I_2 > Br_2 > Cl_2 > F_2$ d) $Cl_2 > Br_2 > F_2 > I_2$

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8) The number of moles of acidified $KMNO_4$ required to oxidize 1 mole of ferrous oxalate (FeC_2O_4) is

a) 5 b) 3 c) 0.6 d) 1.5

9) The pH of 10^{-5} M KOH solution will be

a) 9 b) 5 c) 19 d) none of these

10) Chromyl chloride when dissolved in NaoH solution gives yellow solution. The yellow solution contains.

a) $Cr_2O_7^{2-}$ b) CrO_4^{2-} c) CrO_5 d) Cr_2O_3

11) Which is used in match boxes?

a) White phosphorous b) Red phosphorous

c) Black phosphorus d) Scarlet phosphorous

12) The half life period of a first order reaction is 5 minutes, the time required for 99.9% completeion is nearly equal to

a) 99.9 minutes b) 49.95 minutes c) 50 minutes d) 10 minutes

13) **Match the following:**

1) Lewis acid	-	i) H^+ donor in water
2) Lowry Bronsted theory	-	ii) $K_a = \alpha^2 C$
3) Arrhanius theory	-	iii) Carbonium ion
4) Oswald's dilution law	-	iv) Proton donor

a) 1 - (iii), 2 - (iv), 3 - (i), 4 - (ii)

b) 1 - (iv), 2 - (iii), 3 - (ii), 4 - (i)

c) 1 - (ii), 2 - (iv), 3 - (i), 4 - (iii)

d) 1 - (i), 2 - (iv), 3 - (iii), 4 - (ii)

14) Isopropyl benzene on air oxidation in the presence of dilute acid gives

a) C_6H_5COOH b) $C_6H_5COCH_3$ c) $C_6H_5COC_6H_5$ d) C_6H_5OH

15) Which one of the following is an example of disproportionation reaction?

a) Aldol condensation b) Cannizaro reaction

c) Benzoin condensation d) None of these

PART - II

II. Answer any six questions: 6×2=12

(Question No. 24 is compulsory)

16) Give the basic requirement for vapour phase refining.

17) How will you convert boric acid to boron nitride?

18) Which is more stable? Fe^{3+} or Fe^{2+} explain?

19) Aluminium crystallizes in a cubic close packed structure. Its mettalic radius is 125pm. Calculate the edge length of the unit cell.

20) Write the rate law for the following reactions:

a) A reaction that is $\frac{3}{2}$ order in x and zero order in y.

b) A reaction that is second order in NO and first order in Br_2.

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21) Define ionic product of water. Give the value of room temperature.

22) What is Bayer's reagent? How will you convert ethene to ethane 1,2-diol?

23) What is transesterification reaction?

24) Write a note on dehydration reaction of glycerol.

PART - III

III. Answer any six questions: **6×3=18**

(Question No.33 is compulsory)

25) Write about Magnetic Seperation.

26) **Complete the following reactions:**

i) $P_4 + NaOH + H_2O \rightarrow$

ii) $AgNO_3 + PH_3 \rightarrow$

iii) $KClO_3 \rightarrow$

27) Calculate the number of unpaired electrons and magnetic moment in V^{3+} and Fe^{2+} ions.

28) Write a note on Frankel defect.

29) Explain Common ion effect with an example.

30) Explain Riemer - Tiemann Reaction.

31) Write the test to differentiate ethanol and phenol.

32) How will you convert?

i) Benzene → Acetophenone

ii) Benzaldehyde → Hydrobenzamide

33) Show that for a first order reaction the time required for 99% completion of the reaction is twice the time required for 90% completion of the reaction.

PART - IV

Answer all the questions: **5×5=25**

34) a) i) Explain froth floatation method? **(3m)**

ii) What is cementation? **(2m)**

(OR)

b) i) How will you prepare Borax? **(3m)**

ii) Give the uses of silicones. **(2m)**

35) a) i) Give the equation for the reaction between chlorine with cold NaOH and hot NaOH. 43 G **(3m)**

ii) HF is a weak acid, whereas all other halogen acids are strong acid. Why? 44 G **(2m)**

(OR)

b) i) What is Lanthanoid contraction and what are the effects of Lanthanoid contraction? **(3m)**

ii) Why do transition element form more number of complexes? **(2m)**

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36) a) Calculate the percentage efficiency of packing in case of a face centered cubic crystal? **(5m)**

(OR)

b) Derive integrated rate law for a first order reaction. **(5m)**

37) a) i) Explain Buffer action. **(3m)**

ii) Define Oswald's Dilution Law. **(2m)**

(OR)

b) Differentiate primary, secondary and tertiary alcohols by Victor Meyer's method. **(5m)**

38) a) **Write a note on:** **(5m)**

i) Schotten - Baumann reaction

ii) Knovengal reaction

iii) Cannizaro reaction

(OR)

b) An organic compounds (A) - $C_3H_8O_3$ used as sweetening agent, which on oxidation with Fenton's reagent gives a mixture of compounds B and C. Identify A, B and C. Write possible reactions. **(5m)**

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